NETWORK LAYER DESIGN ISSUES

Network Layer: Design issues ,Routing algorithms: shortest path routing, Flooding, Hierarchical routing, Broadcast, Multicast, distance vector routing, Link State Routing, Congestion Control Algorithms: Approaches to Congestion Control, Traffic aware routing, Admission Control, Traffic throttling, Load shedding.

Design issues:

They explain how packets are forwarded through the network **and** what type of service the network layer provides to the transport layer**. S**ome of the issues that the designers of the network layer must grapple with:

1. Store-and-Forward Packet Switching
2. Services Provided to the Transport Layer
3. Implementation of Connectionless Service
4. Implementation of Connection-Oriented Service

1. Store-and-Forward Packet Switching

In store-and-forward packet switching, a router must receive the entire packet and store it temporarily before forwarding it to the next router. Advantages of this method is Allows routers to check the packet before forwarding. Handles differences in transmission speeds. Packets can be temporarily buffered when the outgoing link is busy.

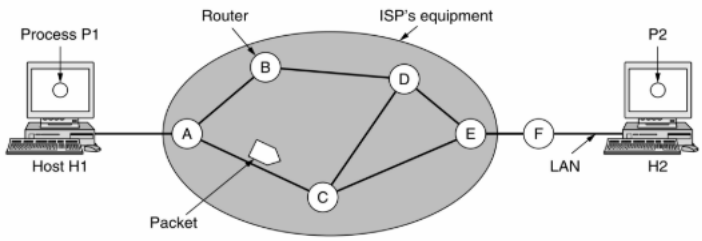


Figure 5-1. The environment of the network layer protocols.

In the above fig A host with a packet to send transmits it to the nearest router, either on its own LAN or over a point-to-point link to the ISP. The packet is stored there until it has fully arrived and the link has finished its processing by verifying the checksum. Then it is forwarded to the next router along the path until it reaches the destination host, where it is delivered. This mechanism is store-and-forward packet switching.

Services Provided to the Transport Layer

The network layer basically provides a way for packets to travel from the source host network to the destination host network. The services need to be carefully designed with the following goals in mind:

1. The services should be independent of the router technology.
2. The transport layer should be shielded from the number, type, and topology of the routers present.
3. The network addresses made available to the transport layer should use a uniform numbering plan, even across LANs and WANs.

There are two major types of service:

1. Connectionless service Also called Datagram service

2. Connection-oriented service Also called Virtual-circuit service

3. Implementation of Connectionless Service (Datagram network):

In a connectionless service, packets are sent individually, and each packet is routed independently. No connection needs to be established before sending data. These packets are frequently called datagrams**,** and the network is called a datagram network. Each packet carries a destination IP address, which routers use to determine where to forward the packet. A router maintains a routing table**,** which helps it forward packets toward their destination. The routing algorithm is responsible for calculating and maintaining routing information and making routing decisions.

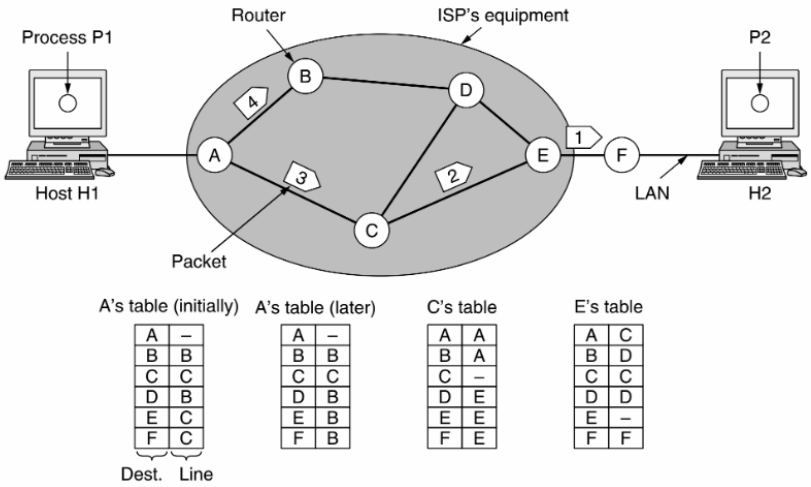


Figure 5-2. Routing within a datagram network.

In the above example At A, packets 1, 2, and 3 are stored briefly, having arrived on the incoming link and had their checksums verified. Then each packet is forwarded according to A's table, onto the outgoing link to C within a new frame. Packet 1 is then forwarded to E and then to F. When it gets to F, it is sent within a frame over the LAN to H2. Packets 2 and 3 follow the same route.

4. Implementation of Connection-Oriented Service

In a **connection-oriented service**, a path is established between the source and destination before any data packets are sent. This connection is called a **Virtual Circuit (VC)**, and the network is called a **virtual-circuit network**. Once the virtual circuit is established, all packets belonging to the same connection generally follow the **same path** from source to destination. Routers maintain information about the virtual circuit, which helps them forward the packets along the established path. The **routing algorithm** is used to determine the suitable path for establishing the virtual circuit.

Example: Here, host H1 has established connection 1 with host H2. This connection is remembered as the first entry in each of the routing tables. The first line of A's table says that if a packet bearing connection identifier 1 comes in from H1, it is to be sent to router C and given connection identifier 1. Similarly, the first entry at C routes the packet to E, also with connection identifier 1. Now let us consider what happens if H3 also wants to establish a connection to H2. It chooses connection identifier 1 (because it is initiating the connection and this is its only connection) and tells the network to establish the virtual circuit.

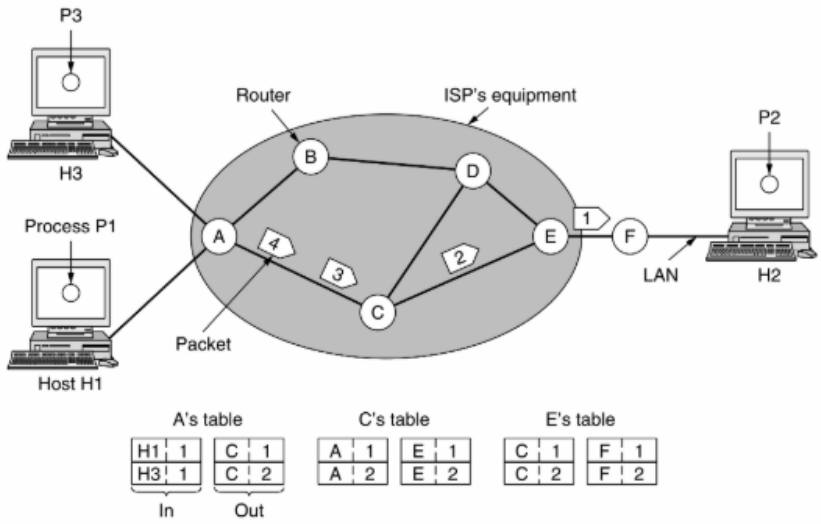


Figure 5-3. Routing within a virtual-circuit network.

This leads to the second row in the tables. Note that we have a conflict here because although A can easily distinguish connection 1 packets from H1 from connection 1 packets from H3, C cannot do this. For this reason, A assigns a different connection identifier to the outgoing traffic for the second connection. Avoiding conflicts of this kind is why routers need the ability to replace connection identifiers in outgoing packets.

Label Switching

Label switching is a technique used in connection-oriented network**s** to forward packets using a short label instead of examining the complete destination IP address at every router. An example of a connection-oriented network service is **MPLS (MultiProtocol Label Switching)**. It is used within ISP networks in the Internet,

Working:

1. A virtual circuit/path is established between the source and destination.
2. A label is assigned to the packets belonging to that connection.
3. When a packet reaches a router, the router looks at the label and uses its forwarding table to decide the next hop.
4. The router may replace the incoming label with a new outgoing label before forwarding the packet.

Comparison of Virtual-Circuit and Datagram Networks

Issue	Datagram network	Virtual-circuit network
Circuit setup	Not needed	Required
Addressing	Each packet contains the full source and destination address	Each packet contains a short VC number
State information	Routers do not hold state information about connections	Each VC requires router table space per connection
Routing	Each packet is routed independently	Route chosen when VC is set up; all packets follow it
Effect of router failures	None, except for packets lost during the crash	All VCs that passed through the failed router are terminated
Quality of service	Difficult	Easy if enough resources can be allocated in advance for each VC
Congestion control	Difficult	Easy if enough resources can be allocated in advance for each VC

1. In a connection-oriented network, what is established before data transfer?

A. MAC address
B. Virtual Circuit
C. DNS connection
D. IP address
Answer: B. Virtual Circuit

2. A connection-oriented network is also called a:

A. Datagram network
B. Broadcast network
C. Virtual-circuit network
D. Local network
Answer: C. Virtual-circuit network

3. In a virtual-circuit network, packets generally:

A. Take random paths
B. Follow the established path
C. Do not contain any identifier
D. Are always broadcast
Answer: B. Follow the established path

4. What is used to identify a connection in label switching?

A. Port number
B. MAC address
C. Connection identifier/label
D. Host name
Answer: C. Connection identifier/label

5. Label switching allows routers to:

A. Forward packets using labels
B. Eliminate all routing tables
C. Avoid establishing connections
D. Change the source host
Answer: A. Forward packets using labels

1. A router uses its _____ table to determine how to forward a packet.

Answer: forwarding / routing

2. A short identifier used for forwarding packets is commonly called a _____.

Answer: label

3. In connectionless service, packets are called _____.

Answer: datagrams

4. In connection-oriented service, the connection is established _____ data transmission.

Answer: before

5. A connection-oriented network is called a _____ network.

Answer: Virtual-circuit